

MILO: An LLM Multi-Stage Conversational Agent for Fostering Teenagers' Mental Resilience

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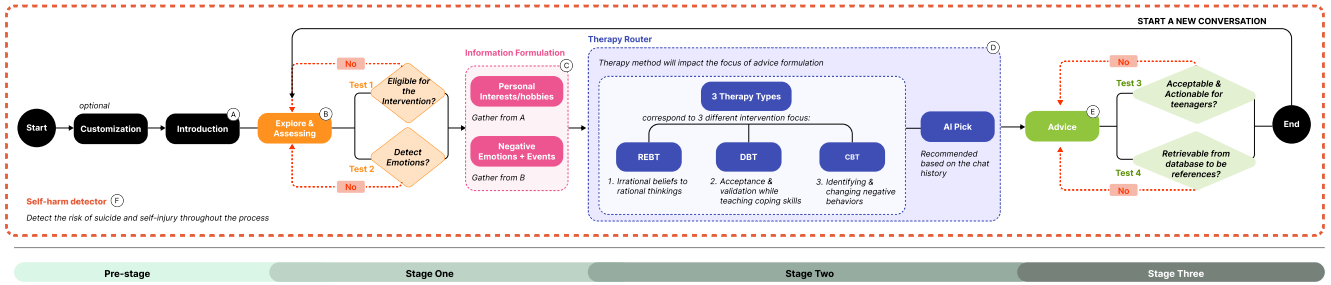


Figure 1: The outline of Milo's LLM conversational logic presents a four-stage CBT workflow, featuring an alert agent for self-harm detection and a therapy router that provides tailored advice to help teenagers self-manage their stress and emotions.

Abstract

Adolescence is a significant period that shapes long-term development and well-being. Mental disorders contribute to 15% of the global disease burden among teenagers, according to the WHO. Adverse well-being during adolescence can not only compromise physical health but also lead to a wide range of negative social outcomes throughout life. Motivated by the potential of generative AI conversational agents to provide scalable and personalized support to cultivate mental resilience, we designed Milo, an LLM digital companion grounded in cognitive behavioral therapy (CBT), tailored specifically for teenagers. Milo promotes greater involvement of teenagers in the development of emotional awareness and resilience strategies through agent customization and offering an interactive interface.

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758411>

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

Conversational Agent, Mental Resilience, Large Language Models

ACM Reference Format:

Han Bao, Yongan Yu, Bohan Wang, Xiaowen Lu, and Xin Tong. 2025. MILO: An LLM Multi-Stage Conversational Agent for Fostering Teenagers' Mental Resilience. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758411>

1 Introduction

Adolescence is a critical life transition from childhood to adulthood, profoundly shaping an individual's future growth and overall well-being; however, teenagers also face significant risks of mental health problems [6]. Approximately 50% of mental health problems begin by age 14 [11]. Adverse well-being during adolescence can not only compromise physical health but also lead to a wide range of negative social outcomes throughout life, including difficulties in interpersonal relationships, lower educational attainment, and limited employment opportunities [2]. Mental resilience refers to the capacity to regulate negative emotions and develop effective coping strategies for stress or adversity [10]. Prior studies have

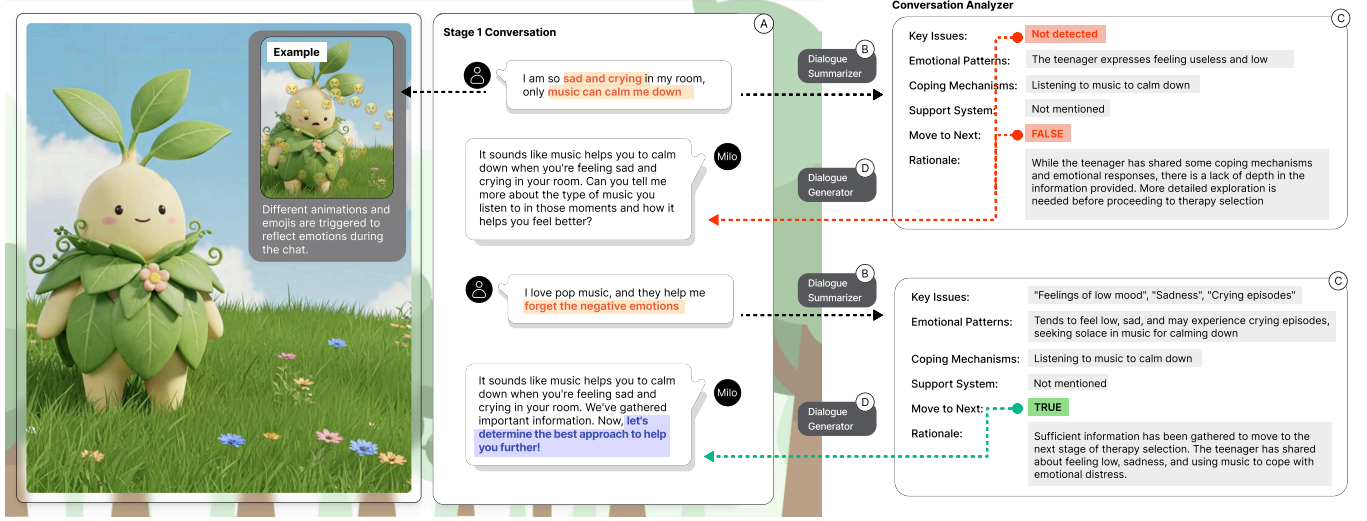


Figure 2: The left panel shows Milo’s interface with an animated avatar and emotional expressions to enhance user engagement; The right panel shows the LLM-guided conversation progresses through multiple turns based on stage-aware logic.

shown that resilience can protect teenagers struggling with depression and anxiety, serving as a buffer against the potential harm of adversity and stressful environments [13]. However, traditional mental resilience training, such as individual counseling or group therapy, faces scalability issues due to cost and space limitations [1]. AI conversational agents (CAs) show great potential in supporting teenagers in facilitating mental resilience. Many studies have applied CAs in psychological domains, demonstrating positive impacts on adult mental health and well-being [3, 4, 8, 9]. Kim et al. [12] indicated that teenagers tend to perceive chatbots as safer listeners for sharing emotions due to their non-judgmental nature. Hu et al. [7] designed an LLM-based conversational agent to promote children’s mental resilience, demonstrating its important role in promoting emotional well-being. Despite these, research and design efforts focused specifically on CAs for teenagers, as well as mental resilience cultivation, remain limited.

Informed by interviews with three professional psychological therapists and an online survey of teenagers ($n = 25$, study protocol approved by the authors’ University IRB), we found that teenagers have limited knowledge of how to build mental resilience by themselves. However, most show a strong willingness to engage with CAs for self-management support. They expressed clear preferences for peer-like, animated cartoon-style agents and valued customization features that promote a more personalized and relatable experience. According to their preferences, we designed Milo, an LLM-based, multi-stage dialogue system grounded in CBT principles. It integrates a mixed-therapy routing mechanism and a retrieval-augmented generation (RAG) pipeline to deliver personalized resilience-building strategies, while learning teenagers’ interests and hobbies [5]. Milo enables teenagers to customize characters by selecting tone, gender, and personality, responding with contextual animations and emojis (Fig. 2) to foster a peer-like atmosphere that supports emotional expression. Through these interactions, teenagers gradually learn to recognize their emotions,

manage negative experiences, build mental resilience, and support long-term well-being.

2 SYSTEM IMPLEMENTATION

Milo integrates multiple LLM components to deliver therapeutic dialogues. GPT-4o manages state transitions and response generation to maintain natural flow, while GPT-3.5-turbo handles summarization, therapy routing, and safety monitoring with lower latency. Safety monitoring includes a self-harm detector that assigns a harm score to each utterance and triggers alerts when multiple risky statements are detected. The system architecture is organized in four key components:

Conversational State Machine. The dialogue is modeled as a four-stage state machine $S = s_p, s_1, s_2, s_3$: *User Customization*, *Information Formulation*, *Therapy Implementation*, and *Advice Intervention*. Transitions between stages are governed by criteria-based functions $T(s_i \rightarrow s_{i+1})$, activated when all stage-specific conditions $C_i = c_1, \dots, c_n$ are met. As such, this structure defines explicit objectives for each stage and guides the dialogue toward targeted therapeutic outcomes.

Dialogue Framework. As illustrated in Figure 2 C, Milo operates two LLM-based agents: a *Dialogue Summarizer* that parses user inputs into structured JSON-based criteria, and a *Dialogue Generator* that selects prompts based on unmet conditions. Unlike static prompting strategies, our stage-aware design enables the system to dynamically adapt to the user’s emotional and informational state. If the “Move To Next” flag is False, the system issues follow-up questions to uncover missing elements, e.g., emotions or coping strategies, to fulfill the requirement for the stage transition.

Therapy Routing Mechanism. During stage s_2 , Milo uses few-shot prompting on prototypical inputs to route users to one of three therapeutic strategies: CBT, dialectical behavior therapy (DBT), or rational emotive behavior therapy (REBT). Each therapy emphasizes different cognitive reframing techniques, as shown in

Figure 1 D). The selected path informs the structure of personalized resilience-building suggestions in later stages.

Advice Generation with RAG. In stage s_3 , Milo employs a RAG module to retrieve advice from a curated corpus [14], incorporating the therapeutic strategy identified in the prior stage. It presents practical tips for teenagers to self-manage their emotions, e.g., exercising, practicing meditation, sourced from trusted health organizations. These entries, tailored to the user's earlier emotional disclosures, provide accessible and actionable guidance that supports mental resilience without being overwhelming.

3 Limitation

While Milo is designed as a conversational agent to support teenagers' mental well-being and resilience, it is not intended to replace professional psychological care and has not yet undergone clinical validation. Future work will include collaboration with licensed psychologists and user studies with teens to rigorously evaluate its safety and effectiveness.

4 Conclusion

This research introduces Milo, an LLM-based conversational agent designed to help teenagers build mental resilience. Tailored for teenagers, Milo features a customizable, cartoon-style character with interactive animations and emojis. Grounded in CBT and powered by a RAG pipeline, it delivers personalized self-support strategies through a friendly, chat-like experience that encourages emotional reflection and resilience-building.

Acknowledgments

Han Bao and Yongan Yu contributed to this work as research assistants at HKUST (GZ). We gratefully acknowledge the support of the Guangzhou City-University Joint Research Grant (Project No. GP2025M024, Grant No. P00437), funded by the Guangzhou Science and Technology Bureau, and the WZ Project, funded by the State Administration of Foreign Experts Affairs (SAFEA) (Grant No. G01MF010061), for their contributions to this research.

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